



Submittal Review Response

Project Name: Hilo WWTP Rehabilitation and Replacement Project Phase 1
Submittal No.: 05500-001.1
Date: 9/30/2025

Client: County of Hawai'i **Carollo Project No.:** 203975
Contractor: Nan, Inc.
Submittal Name: Grating Support in Grit Basin - Shop Drawing Resubmittal
Reviewed By: Marissa Kurniawan, Hipom Caleb Che

SUBMITTAL REVIEW

Review is for general compliance with contract documents. No responsibility is assumed by Carollo for correctness of quantities, dimensions, and details. No deviation or variation is approved unless specifically addressed in these review comments. Refer to Section 01330 for additional requirements. The Contractor shall assume full responsibility for coordination with all other trades and deviations from contract requirements.

Approved	☐	No Exceptions
	☐	Make Corrections Noted - See Comments
	☒	Make Corrections Noted - Confirm
Not Approved	☐	Correct and Resubmit
	☐	Rejected - See Remarks
Receipt Acknowledged	☐	Filed for Record
	☐	With Comments - Resubmit

Review Comments:

Previous Comments:

- Refer to attached PDF for comments. Verify and address all comments.
 - Contractor's Response: Acknowledged
 - Engineer's Response: Acknowledged. Comment Closed.
- Contractor to verify and confirm all clouded dimensions.
 - Contractor's Response: Acknowledged
 - Engineer's Response: Acknowledged. Comment Closed.
- Per specification 05500-3.02.A.2., Contractor shall coat all aluminum surfaces in contact with concrete.
 - Contractor's Response: Incorporated in Page 13 and Page 14.
 - Engineer's Response: Acknowledged. Comment Closed.
- The submitted shop drawing appears to not specify aluminum alloy grade. Provide 6061-T6 or 6063-T6 per specification 05500-2.02.H.2.
 - Contractor's Response: Incorporated in Page 13 and Page 14.
 - Engineer's Response: Acknowledged. Comment Closed.
- Do not use bent plates for aluminum ledger support. Bending of aluminum plate may lead to cracking and breaking.

Provide AL L3x2x1/4 per Detail S542/Typ for the ledger support.

- a. Contractor's Response: Incorporated in Page 12.
 - b. Engineer's Response: Acknowledged. Comment Closed.
6. Ledger angle concrete anchors shall be Type 316 SST per S542/TYP.
- a. Contractor's Response: Incorporated in Page 12.
 - b. Engineer's Response:
 - 1) Shop drawings show SST but not the grade of the anchor. Confirm that ledger angle concrete anchors are Type 316.
 - 2) Do not interrupt reinforcing bars for installation of post-installed anchors.
7. Provide welding details of rebate per S542/TYP.
- a. Contractor's Response: Incorporated in Page 12 and Page 13.
 - b. Engineer's Response: Acknowledged. Comment Closed.

New Comments:

- 8. All clouded sizes/dimensions shown are confirmed. Contractor to field verify and coordinate all final dimensions.
- 9. Submittal page 13 calls out cross section B-B, but B-B is not shown anywhere in the submittal.

CONTRACTOR SUBMITTAL TRANSMITTAL FORM REV. A

Owner: County of Hawaii
Contractor: Nan, Inc. **Project No.:** WW-4705R
Project Name: Hilo WWTP Phase 1 **Submittal Number:**
Submittal Title: For Information Only
TO:
From: Nan Inc.

Specification No. and Subject of Submittal / Equipment Supplier	
Spec:	Paragraph:
Authored By:	Date Submitted:

Submittal Certification	
Check Either (A) or (B):	
☐ (A)	We have verified that the equipment or material contained in this submittal meets all the requirements specified in the project manual or shown on the contract drawings with <u>no exceptions</u> .
☐ (B)	We have verified that the equipment or material contained in this submittal meets all the requirements specified in the project manual or shown on the contract drawings <u>except</u> for the deviations listed.
Certification Statement: By this submittal, I hereby represent that I have determined and verified all field measurements, field construction criteria, materials, dimensions, catalog numbers and similar data, and I have checked and coordinated each item with other applicable approved shop drawings and all Contract requirements.	
General Contractor's Reviewer's Signature: 	
Printed Name and Title:	
In the event, Contractor believes the Submittal response does or will cause a change to the requirements of the Contract, Contractor shall immediately give written notice stating that Contractor considers the response to be a Change Order.	
Firm:	Signature: Date Returned:

PM/CM Office Use
Date Received GC to PM/CM:
Date Received PM/CM to Reviewer:
Date Received Reviewer to PM/CM:
Date Sent PM/CM to GC:

Nan, Inc

PROJECT: HILO WWTP REHABILITATION
AND REPLACEMENT PROJECT - PHASE 1

JOB NO. WW-4705R

THIS SUBMITTAL HAS BEEN CHECKED BY
THIS CONTRACTOR. IT IS CERTIFIED
CORRECT, COMPLETE, AND IN
COMPLIANCE WITH CONTRACT
DRAWINGS AND SPECIFICATIONS. ALL
AFFECTED CONTRACTORS AND
SUPPLIERS ARE AWARE OF, AND WILL
INTEGRATE THIS SUBMITTAL (UPON
APPROVAL) INTO THEIR OWN WORK.

DATE RECEIVED _____
 SPECIFICATION SECTION # _____
 SPECIFICATION _____
 PARAGRAPH _____
 DRAWING _____
 SUBCONTRACTOR _____
 SUPPLIER _____
 MANUFACTURER _____

CERTIFIED BY CQCM or Designee :


 Matthew Chan, QC Engineer

Review Comments:

1. Refer to attached PDF for comments. Verify and address all comments.
[Acknowledged.](#)
2. Contractor to verify and confirm all clouded dimensions.
[Acknowledged.](#)
3. Per specification 05500-3.02.A.2., Contractor shall coat all aluminum surfaces in contact with concrete.
[Incorporated in page 13 and page 14.](#)
4. The submitted shop drawing appears to not specify aluminum alloy grade. Provide 6061-T6 or 6063-T6 per specification 05500-2.02.H.2.
[Incorporated in page 13 and page 14.](#)
5. Do not use bent plates for aluminum ledger support. Bending of aluminum plate may lead to cracking and breaking. Provide AL L3x2x1/4 per Detail S542/Typ for the ledger support.
[Incorporated in page 12.](#)
6. Ledger angle concrete anchors shall be Type 316 SST per S542/TYP.
[Incorporated in page 12.](#)
7. Provide welding details of rebate per S542/TYP.
[Incorporated in page 12 and page 13.](#)

2.02 MANUFACTURED UNITS

- A. Aluminum grating stair tread:
 - 1. Manufacturers: One of the following or equal:
 - a. Harsco Industrial IKG, Aluminum Grating Stair Tread with Mebac® nosing.
 - b. McNichols Co., Type A-Standard with Corrugated Angle Nosing.
 - 2. Material: Welded aluminum grating tread with non-slip nosing and integral end plates for bolt on attachment to stair stringers.
 - 3. Size:
 - a. Tread width: To equal tread spacing plus 1 inch minimum.
 - b. Tread length: Length to suit stringer-to-stringer dimension on the Drawings.
 - c. Depth: 1-3/4 inches.
 - 4. Bolts: Type 316 stainless steel.
- B. Aluminum stair nosing:
 - 1. Manufacturers: One of the following or equal:
 - a. Wooster Products, Inc., Type 101 Nosing.
 - b. American Safety Tread Co., Inc., Style 801 Nosing.
 - 2. Material: Cast aluminum abrasive nosings with aluminum oxide granules integrally cast into metal, forming permanent, nonslip, long-wearing surface.
 - 3. For installation in cast-in-place stairs.
 - 4. Configuration: 4 inches wide, fabricated with integrally cast stainless steel anchors at approximately 12-inch centers. Length to extend within 3 inches of stair edge on each side.
- C. Cast iron stop plank grooves:
 - 1. Manufacturers: One of the following or equal:
 - a. Neenah Foundry Co., R-7500 Series, Type A.
 - b. McKinley Iron Works, Type L.
 - 2. Size: 2-inch wide groove opening by 1-1/2 inch deep, unless otherwise indicated on the Drawings.
 - 3. Recess groove with the cast iron surface of the groove set flush with the concrete surface.
- D. Concrete inserts:
 - 1. Concrete inserts for supporting pipe and other applications are specified in Section 15061 - Pipe Supports.
- E. Handrails and guardrails:
 - 1. General:
 - a. Design and fabricate assemblies to conform to current local, State, OSHA and HIOSH standards and requirements.
 - b. Coordinate layout of assemblies and post spacings to avoid conflicts with equipment and equipment operators:
 - 1) Indicate on the shop drawings locations of such equipment.
 - 2) Highlight locations where railings cannot be made continuous, and obtain Engineer's directions on how to proceed before fabricating or installing railings.

2. Aluminum handrails and guardrails (nonwelded pipe):
 - a. Rails, posts, and fitting-assembly spacers:
 - 1) In accordance with ASTM B429, 6005, 6063 or 6105, minimum Schedule 40, extruded aluminum pipe of minimum 1.89-inch outside diameter and 0.14-inch wall thickness.
 - b. Kick plates: 6061 or 6105 aluminum alloy.
 - c. Fastenings and fasteners: As recommended or furnished by the manufacturer.
 - d. Other parts: 6063 extruded aluminum, or F214 or F514.0 aluminum castings:
 - 1) Fabrications: In accordance with ASTM B209 or ASTM B221 extruded bars:
 - a) Bases: 6061 or 6063 extruded aluminum alloy.
 - 2) Plug screws or blind rivets: Type 305 stainless steel.
 - a) Other parts: Type 300 series stainless steel.
 - e. Finish of aluminum components:
 - 1) Anodized finish, 0.7 mil thick, applied to exposed surfaces after cutting. Aluminum Association Specification M12-C22-A41, mechanical finish non specular as fabricated, chemical finish-medium matte, anodic coating-clear Class I Architectural.
 - 2) Pretreat aluminum for cleaning and removing markings before anodizing.
 - f. Fabrication and assembly:
 - 1) Fabricate posts in single, unspliced pipe length.
 - 2) Perform without welding.
 - 3) Do not epoxy bond the parts.
 - 4) Maximum clear opening between assembled railing components as indicated on the Drawings.
 - g. Manufacturers: One of the following or equal:
 - 1) Moultrie Manufacturing Co., Wesrail.
 - 2) Golden Railings, Riveted System.
 - 3) Craneveyor Corp. Enerco Metals, C-V Rail.
3. Guardrail gates:
 - a. Supplied by guardrail manufacturer:
 - 1) Of same material, quality, and workmanship as specified for guardrail system in which they will be installed.
 - 2) Of design similar to that of handrail or railing system in which they will be installed.
 - b. Components: Gate frame, stainless steel self-closing device, hinges, gate stops, and durable self-locking type latch. Fabricate components in conformance with OSHA and HIOSH minimum strength requirements.
4. Fastenings and fasteners: As recommended or furnished by guardrail manufacturer for use with this system.

F. Ladders:

1. General:
 - a. Type: Safety type conforming to local, State, OSHA and HIOSH standards as minimum. Furnish guards for ladder wells.
 - b. Size: 18 inches wide between side rails of length, size, shape, detail, and location indicated on the Drawings.

2. Aluminum ladders:
 - a. Materials: 6063-T5 aluminum alloy.
 - b. Rungs:
 - 1) 1-inch minimum solid square bar with 1/8-inch grooves in top and deeply serrated on all sides.
 - 2) Capable of withstanding 1,000 pound load without failure.
 - c. Side rails: Minimum 4-inch by 1/2-inch flat bars.
 - d. Finish of aluminum components:
 - 1) Anodized finish, 0.7 mil thick, applied to exposed surfaces after cutting. Aluminum Association Specification M12-C22-A41, mechanical finish non specular as fabricated, chemical finish-medium matte, anodic coating-clear Class I Architectural.
 - 2) Pretreat aluminum for cleaning and removing markings before anodizing.
 - e. Fabrication:
 - 1) Welded construction, of size, shape, location, and details indicated on the Drawings.
 - 2) For ladders over 6 feet high, furnish fall prevention system designed in accordance with State, OSHA and HIOSH requirements.
 - f. Fall prevention system: Include but not limit to railing, brackets, clamps, 2 sleeves, and 2 belts, satisfying OSHA and HIOSH safe climbing requirements:
 - 1) Manufacturers: One of the following or equal:
 - a) North Consumer Products, Saf-T-Climb.
 - b) Swager Communications, Climbers Buddy System.
- G. Manhole frames and covers:
 1. Material: Gray iron castings, in accordance with ASTM A48, Class 30-B.
 2. Type: Heavy-duty traffic type, with combined minimum set weight of 265 pounds.
 3. Machine horizontal and vertical bearing surfaces to fit neatly, with easily removable cover bearing firmly in frame without rocking.
 4. Frame:
 - a. Bottom flange type.
 - b. Approximately 4-1/2 inches frame height.
 - c. Dimensions as indicated on the Drawings:
 - 1) Minimum inside clear dimension may not be smaller than nominal diameter minus 2 inches.
 5. Cover:
 - a. Skid-resistant grid pattern design stamped with name of utility service provided by manhole, such as "ELECTRICAL," "SEWER," "TELEPHONE," or "WATER."
 - b. Solid type without ventilation holes.
 6. Finish: Unpainted.
- H. Metal gratings:
 1. General:
 - a. Fabricate grating to cover areas indicated on the Drawings.
 - b. Unless otherwise indicated on the Drawings, grating over an opening shall cover entire opening.

- c. Make cutouts in grating where required for equipment access or protrusion, including valve operators or stems, and gate frames.
- d. Band ends of grating and edges of cutouts in grating:
 - 1) End banding: 1/4 inch less than height of grating, with top of grating and top edge of banding flush.
 - 2) Cutout banding: Full-height of grating.
 - 3) Use banding of same material as grating.
 - 4) Panel layout: Enable installation and subsequent removal of grating around protrusions or piping.
 - 5) Openings 6 inches and larger: Lay out grating panels with edges of 2 adjacent panels located on centerline of opening.
 - 6) Openings smaller than 6 inches: Locate opening at edge of single panel.
 - 7) Where an area requires more than 1 grating section to cover area, clamp adjacent grating sections together at 1/4-points with fasteners acceptable to Engineer.
 - 8) Fabricate steel grating sections in units weighing not more than 50 pounds each.
 - 9) Fabricate aluminum grating sections in units weighing not more than 50 pounds each.
 - 10) Gaps between adjacent grating sections shall not be more than the clear spacing between bearing bars.
- e. When requested by Engineer, test 1 section of each size grating for each span length involved on the job under full load:
 - 1) Furnish a suitable dial gauge for measuring deflections.
- f. Grating shall be aluminum, unless otherwise specified or indicated on the Drawings.

2. Aluminum grating:

- a. Material for gratings, shelf angles, and rebates: 6061-T6 or 6063-T6 aluminum alloy, except crossbars may be 6063-T5 aluminum alloy.
- b. Shelf angle concrete anchors: Type 304 or Type 316 stainless steel.
- c. Grating rebate rod anchors: 6061-T6 or 6063-T6 aluminum alloy.
- d. Bar size and spacing: As determined by manufacturer to enable grating to support design load.
- e. Design live load: A minimum of 100 pounds per square foot uniform live load on entire grating area, but not less than the live load indicated on the Drawings for the area where grating is located.
- f. Maximum fiber stress for design load: 12,000 pounds per square inch.
- g. Maximum deflection due to design load: 1/240 of grating clear span.
- h. Maximum spacing of main grating bars: 1-1/8 inches clear between bars.
- i. Minimum grating height: 1-1/2 inches.
- j. Manufacturers: The following or equal:
 - 1) Harsco Industrial IKG, Swaged Aluminum I-Bar with striated finish.

3. Steel gratings:

- a. Hot-dip galvanized in accordance with ASTM A123.
- b. Bar size and spacing: As determined by the manufacturer to support design load.
- c. Design live load: A minimum of 100 pounds per square foot uniform live load on the entire area of the grating area, but not less than the live load indicated on the Drawings for the area where the grating is located.

- d. Maximum fiber stress for design load: 18,000 pounds per square inch.
 - e. Maximum deflection under design load: 1/240 of grating clear span.
 - f. Bar spacing: Maximum of 1-1/8 inches clear between bars.
 - g. Manufacturers: The following or equal:
 - 1) Harsco Industrial IKG, IKG Weldforged.
- 4. Heavy-duty steel grating:
 - a. Heavy-duty type, fabricated from structural steel and designed in accordance with AASHTO Standard Specifications for Highway Bridges, using H-20 loading.
 - b. Hot-dip galvanized after fabrication in accordance with ASTM A123.
 - c. Manufacturers: One of the following or equal:
 - 1) Reliance Steel Products Co., Heavy-Duty Steel Grating.
 - 2) Seidelhuber Metal Products, Inc., equivalent product.
- I. Metal tread plate:
 - 1. Plate having a raised figured pattern on 1 surface to provide improved traction.
- J. Preformed channel pipe supports:
 - 1. Preformed channel pipe supports for pipe supports and other applications are specified in Section 15062 - Preformed Channel Pipe Support System.
- K. Stairs:
 - 1. Aluminum stairs:
 - a. Stringers: 6061-T6 aluminum alloy.
 - b. Stair treads:
 - 1) Aluminum of same type specified under Aluminum Grating.
 - 2) Of sizes indicated on the Drawings, and 1-3/4 inch minimum depth with cast abrasive type safety nosings.
 - c. Provide a vertical close piece between each riser. Fabricate, install, and fasten close pieces as indicated on the Drawings.
 - d. Handrails and guardrails: Aluminum pipe specified under Aluminum Handrails and Guardrails (Nonwelded Pipe).
 - e. Fasteners: Type 304 or Type 316 stainless steel.
- L. Miscellaneous aluminum:
 - 1. Fabricate aluminum products, not covered separately in this Section, in accordance with the best practices of the trade and field assemble by riveting or bolting.
 - 2. Do not weld or flame cut.
- M. Miscellaneous cast iron:
 - 1. General:
 - a. Tough, gray iron, free from cracks, holes, swells, and cold shuts.
 - b. Quality such that hammer blow will produce indentation on rectangular edge of casting without flaking metal.
 - c. Before leaving the foundry, clean castings and apply 16-mil dry film thickness coating of coal-tar epoxy, unless otherwise specified or indicated on the Drawings.

- N. Miscellaneous stainless steel:
1. Provide miscellaneous stainless steel items not specified in this Section as indicated on the Drawings or specified elsewhere:
 - a. Fabricate and install in accordance with the best practices of the trade.
 2. Cleaning and passivation:
 - a. Following shop fabrication of stainless steel members, clean and passivate fabrications.
 - b. Finish requirements: Remove free iron, heat tint oxides, weld scale and other impurities, and obtain a passive finished surface.
 - c. Provide quality control testing to verify effectiveness of cleaning agents and procedures and to confirm that finished surfaces are clean and passivated:
 - 1) Conduct sample runs using test specimens with proposed cleaning agents and procedures as required to avoid adverse effects on surface finishes and base materials.
 - d. Pre-clean, chemically descale (pickle), and final clean fabrications in accordance with the requirements of ASTM A380 to remove deposited contaminants before shipping:
 - 1) Passivation by citric acid treatment is not allowed.
 - a) If degreasing is required before cleaning to remove scale or iron oxide, cleaning (pickling) treatments with citric acid are permissible; however, these treatments shall be followed by inorganic cleaners such as nitric-hydrofluoric acid.
 - 2) Provide acid descaling (pickling) in accordance with Table A1.1 of Annex A1 of ASTM A380.
 - 3) After pickling, final cleaning of stainless steel shall conform to Part II of Table A2.1 of Annex A2 of ASTM A380.
 - e. After cleaning, inspect using methods specified for "gross inspection" in ASTM A380.
 - f. Improperly or poorly cleaned and passivated materials shall not be shipped and will not be accepted at the job site.
- O. Miscellaneous structural steel:
1. Provide miscellaneous steel items not specified in this Section as indicated on the Drawings or specified elsewhere:
 - a. Fabricate and install in accordance with the best practices of the trade.
- P. Isolating sleeves and washers:
1. As indicated on the Drawings and as specified in Section 05190 - Mechanical Anchoring and Fastening to Concrete and Masonry.

PART 3 EXECUTION

3.01 EXAMINATION

- A. Verification of conditions:
1. Examine work in place to verify that it is satisfactory to receive the work of this Section.
 2. If unsatisfactory conditions exist, do not begin this work until such conditions have been corrected.

3.02 INSTALLATION

A. General:

1. Install products as indicated on the Drawings, and in accordance with shop drawings and manufacturer's printed instructions, as applicable except where specified otherwise.

2. Interface between materials:

- a. Dissimilar metals: Where steel comes in contact with dissimilar metals (aluminum, stainless steel, etc.), separate or isolate the dissimilar metals:
 - 1) Make application so that the isolating or protective barrier is not visible in the completed construction.
 - 2) Isolating sleeves and washers: As specified in Section 05190 - Mechanical Anchoring and Fastening to Concrete and Masonry.
- b. Aluminum in contact with concrete or masonry: Coat aluminum surfaces as specified in Section 09960 - High Performance Coatings.
- c. Aluminum in contact with concrete or masonry.

B. Aluminum stair nosing:

1. Install stair nosings on treads of concrete stairs, including top tread on upper concrete slab.
2. Omit stair nosings where concrete is submerged.
3. Cast stair nosings in fresh concrete, flush with tread and riser faces. Install nosing in center of step.

C. Cast iron stop plank grooves:

1. Recess stop plank grooves with cast iron surfaces of groove set flush with concrete surface.

D. Handrails and guardrails:

1. General:

- a. Fasten pipe rails to fittings with Series 300 stainless steel pop rivets or flush set screws.
- b. Make pipe cuts clean and straight, free of burrs and nicks, and square and accurate for minimum joint-gap.
- c. Drill and countersink holes to proper size, as required for a tight flush fit of screws and other component parts.
- d. Space attachment brackets as indicated in the manufacturer's instructions.

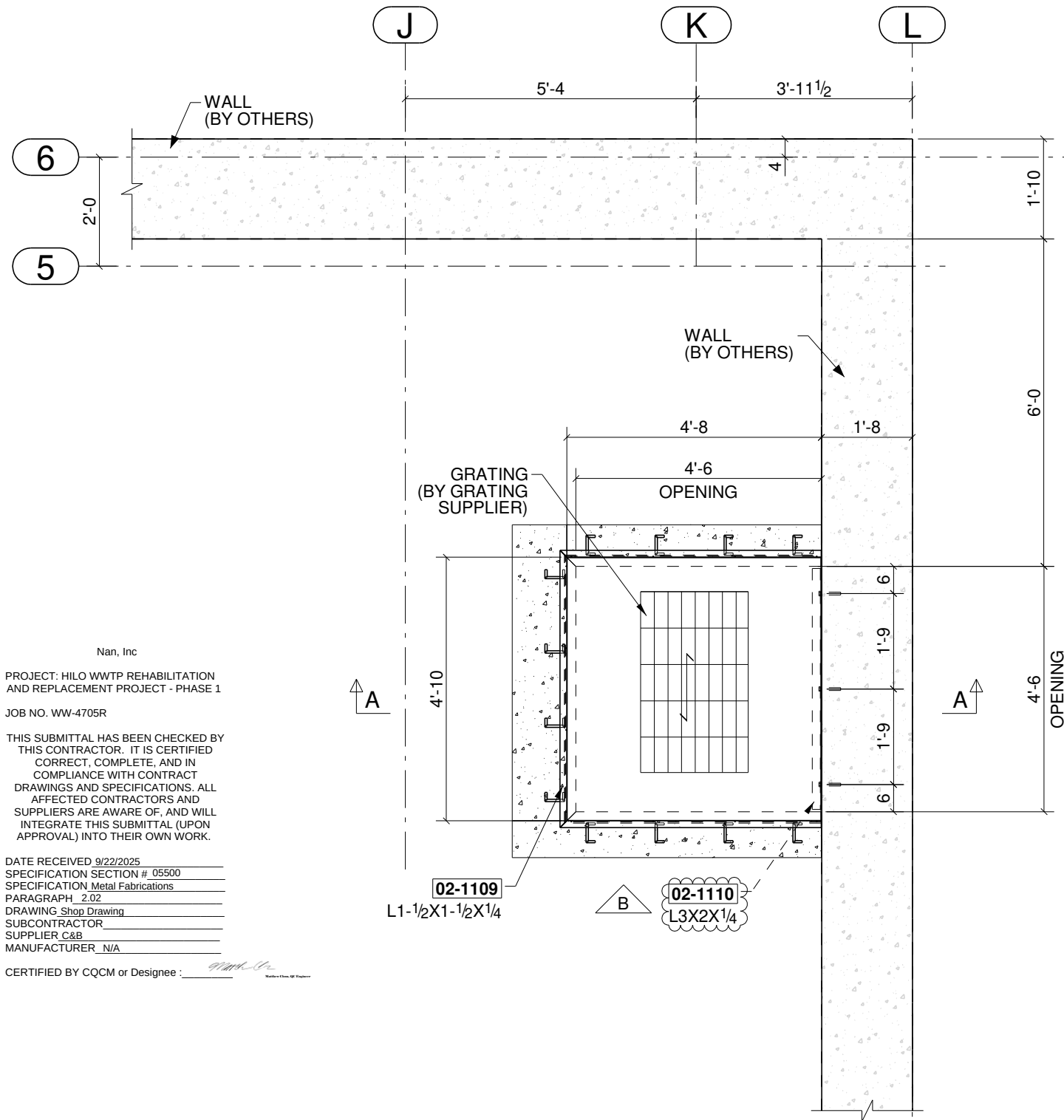
2. Aluminum pipe handrails and guardrails:

- a. During construction, keep exterior surfaces of handrails and guardrails covered with minimum 0.4 millimeters of heat shrink polyethylene film.
- b. Do not remove protective film before handrails and guardrails have been accepted by Engineer nor before other work in proximity of handrails and guardrails has been completed.
- c. Discontinue handrails and guardrails at lighting fixtures.
- d. Provide 1/8-inch diameter weep hole at base of each post.
- e. Space posts as indicated on the Drawings.
- f. Anchor posts into concrete by grouting posts into formed holes in concrete, into stainless steel sleeves cast in concrete; or bracket mount to face of concrete surfaces as specified and indicated on the Drawings.
- g. Space rails as indicated on the Drawings.

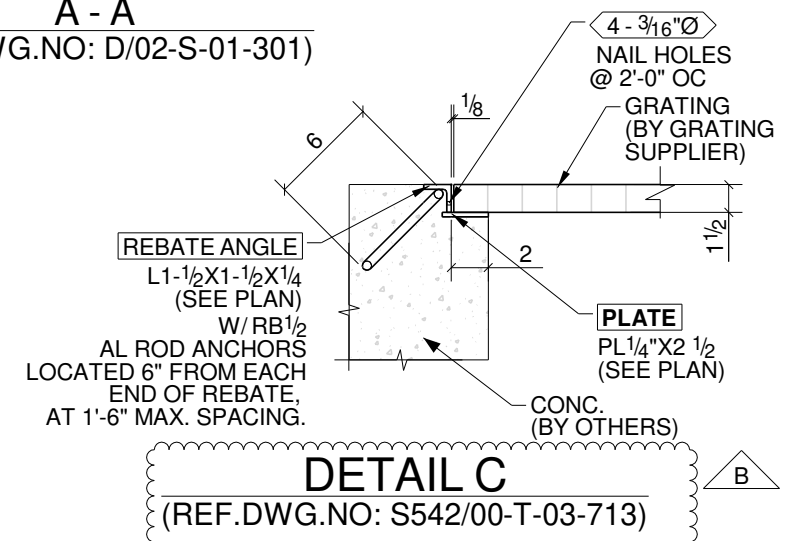
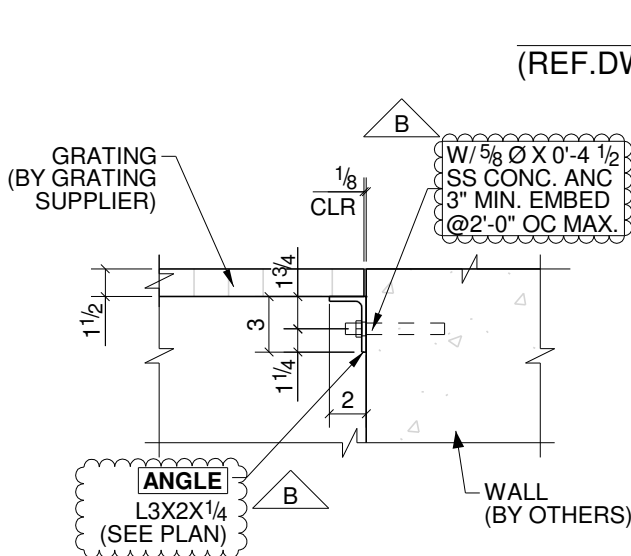
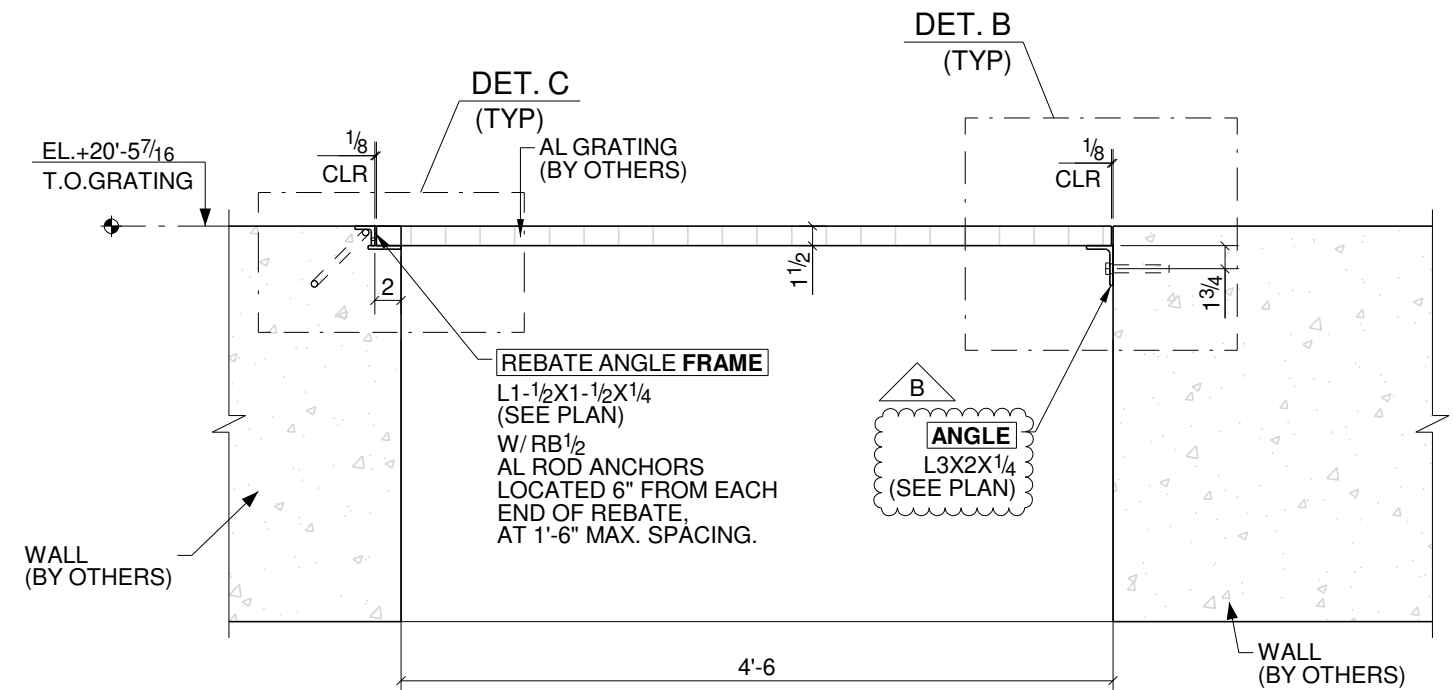
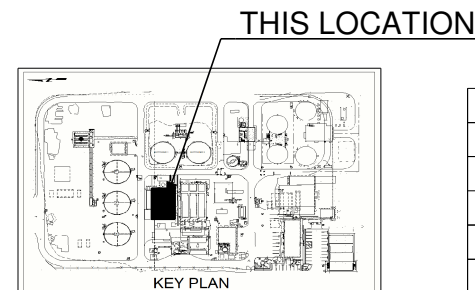
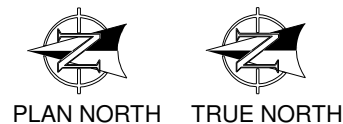
- h. Make adequate provision for expansion and contraction of kick plates and rails:
 - 1) Make provisions for removable sections where indicated on the Drawings.
- i. Make lower rails a single, unspliced length between posts, or continuous.
- j. Make top rails continuous whenever possible, and attach single, unspliced lengths to 3 posts minimum.
- k. Draw up fasteners tight with hand wrench or screw driver.
- l. Space attachment brackets as indicated on shop drawings or in manufacturer's installation instructions.
- m. Completed installation shall have handrails and railings rigid and free of play at joints and attachments.
- n. Protect handrail and guardrail finish from scratches, gouges, dents, stains, and other damage.
- o. Replace damaged or disfigured handrails and guardrails with new.
- p. Shortly before final acceptance of the work, and after removal of protective polyethylene film, clean handrails and guardrails with mild detergent or with soap and water:
 - 1) After cleaning, thoroughly rinse handrails and guardrails and wipe with soft cloth.
- q. Erect guardrail straight, level, plumb, and true to the positions as indicated on the Drawings. Correct deviations from true line of grade, which are visible to the eye.
- 3. Guardrail gates:
 - a. Install gate to be a vertical plane with the guardrail when in the closed position.
 - b. Install hinges so that each gate can swing 180 degrees from the closed position to the fully open position.
 - c. Install so that the gates swing to the walkway side of the guardrail only:
 - 1) Install gate stops on the stationary railing posts to prohibit gates from swinging in the wrong direction.
 - d. Install gate frames, hinges, stops, and latches in conformance with OSHA and HIOSH minimum strength requirements.
- E. Ladders:
 - 1. Secure to supporting surface with bent plate clips providing minimum 8 inches between supporting surface and center of rungs.
 - 2. Where exit from ladder is forward over top rung, extend side rails 3 feet 3 inches minimum above landing, and return the rails with a radius bend to the landing.
 - 3. Where exit from ladder is to side, extend ladder 5 feet 6 inches minimum above landing and rigidly secure at top.
 - 4. Erect rail straight, level, plumb, and true to position indicated on the Drawings:
 - a. Correct deviations from true line or grade which are visible to the eye.
- F. Manhole frames and covers:
 - 1. Installation: As recommended by Manufacturer.

- G. Metal gratings:
 - 1. General:
 - a. Allow 1/8-inch maximum clearance between ends of grating and inside face of vertical leg of shelf angles.
 - b. Horizontal bearing leg of shelf angles shall be 2 inches minimum.
 - c. Install aluminum plate or angles where necessary to fill openings at changes in elevation and at openings between equipment and grating.
 - d. Install angle stops at ends of grating.
 - e. Installed grating shall not slide out of rebate or off support.
 - f. Weld stops in place, unless otherwise specified or indicated on the Drawings.
 - g. Top surfaces of grating sections adjacent to each other shall lie in same plane.
 - 2. Aluminum grating:
 - a. Aluminum grating: Support on aluminum shelf angles or rebates.
 - 3. Steel grating:
 - a. Support on hot-dip galvanized structural steel shelf angles or rebates.
 - 4. Heavy-duty steel grating:
 - a. Support on hot-dip galvanized structural steel rebates embedded and anchored in concrete.
 - b. Use for roadways, traffic areas, and where indicated on the Drawings.
- H. Stairs:
 - 1. General:
 - a. Install guard railings around stair wells as indicated on the Drawings or specified.
- I. Stainless Steel:
 - 1. Welding:
 - a. Passivate field-welded surfaces:
 - 1) Provide cleaning, pickling and passivating as specified in this Section.
 - 2) Clean using Derustit Stainless Steel Cleaner, or equal.

END OF SECTION



ALUMINUM GRATING SUPPORT FRAME PLAN
REF.DWG.=02-S-01-105



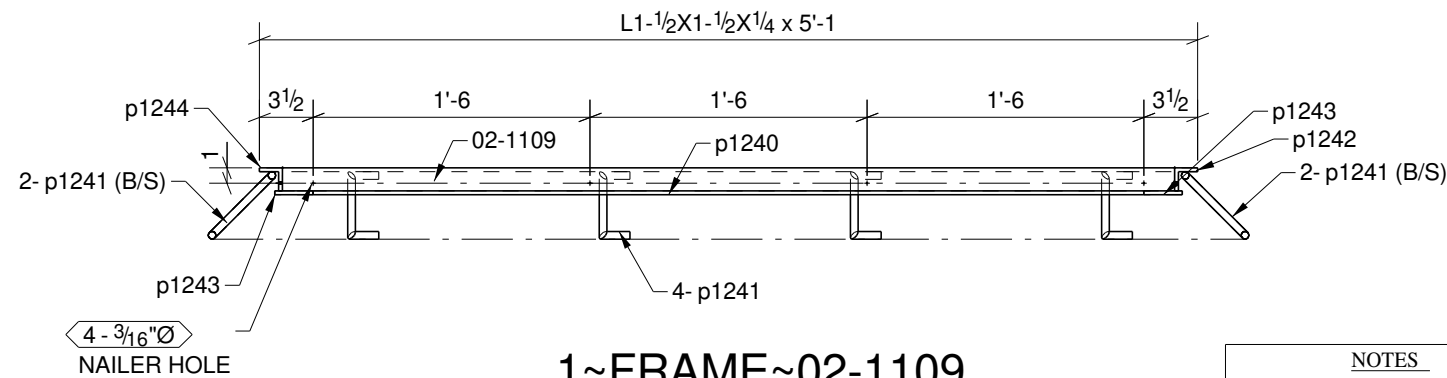
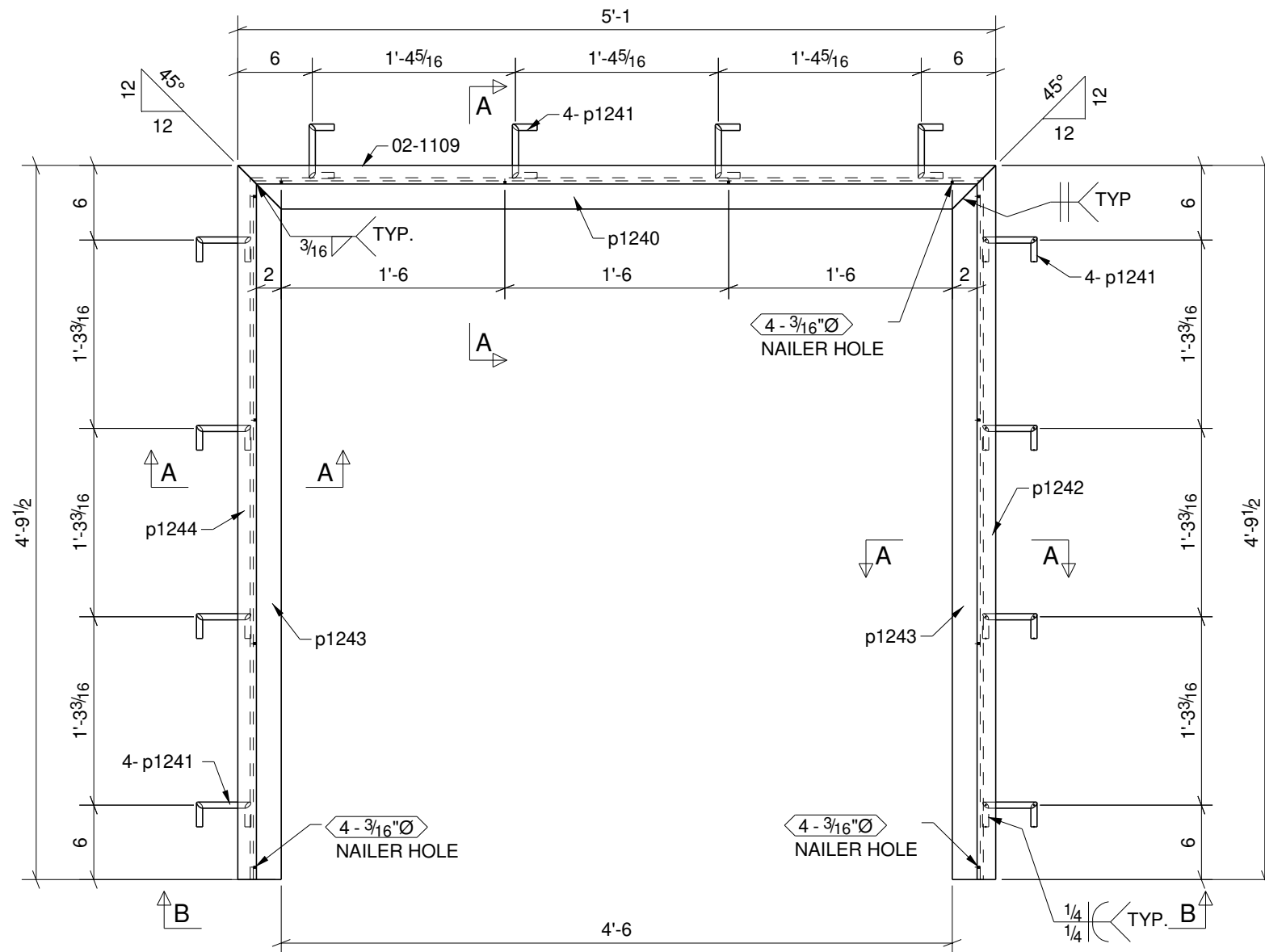
GC NOTE:
ALL DIMENSIONS
NEEDS TO BE
FIELD VERIFIED PRIOR
TO FABRICATION.

HEADWORKS

Nan, Inc
PROJECT: HILO WWTP REHABILITATION
AND REPLACEMENT PROJECT - PHASE 1
JOB NO. WW-4705R
THIS SUBMITTAL HAS BEEN CHECKED BY
THIS CONTRACTOR. IT IS CERTIFIED
CORRECT, COMPLETE, AND IN
COMPLIANCE WITH CONTRACT
DRAWINGS AND SPECIFICATIONS. ALL
AFFECTED CONTRACTORS AND
SUPPLIERS ARE AWARE OF, AND WILL
INTEGRATE THIS SUBMITTAL (UPON
APPROVAL) INTO THEIR OWN WORK.
DATE RECEIVED 9/22/2025
SPECIFICATION SECTION # 05500
SPECIFICATION Metal Fabrications
PARAGRAPH 2.02
DRAWING Shop Drawing
SUBCONTRACTOR
SUPPLIER C&B
MANUFACTURER N/A
CERTIFIED BY CQCM or Designee: *[Signature]*

DETAILER	ABM	CHECKER	KAR	 SPECIALTY METALS FABRICATION 12002 N. LAMAR BLVD. AUSTIN, TEXAS 78753 WWW.CODYBUILDERSSUPPLY.COM (512) 339-9834	
WELDS	1/4 U.N.O.	HOLES	13/16 Ø U.N.O.		
SURF PREP					
PAINT					
REVISIONS				DETAILS OF	ALUMINUM GRATING SUPPORT FRAME PLAN
No.	DATE	BY	DESCRIPTION	CUSTOMER	COUNTRY OF HAWAII
A	06/30/2025	MKV	FOR APPROVAL	PROJECT	HILO WWTP REHABILITATION
B	09/22/2025	DUS	REVISED & RE-SUBMIT FOR APPROVAL	LOCATION	HILO, HAWAII
				ENGINEER	BRIAN F. FUNAI
					JOB No.
					23240832
					DRAWING No.
					02-E139

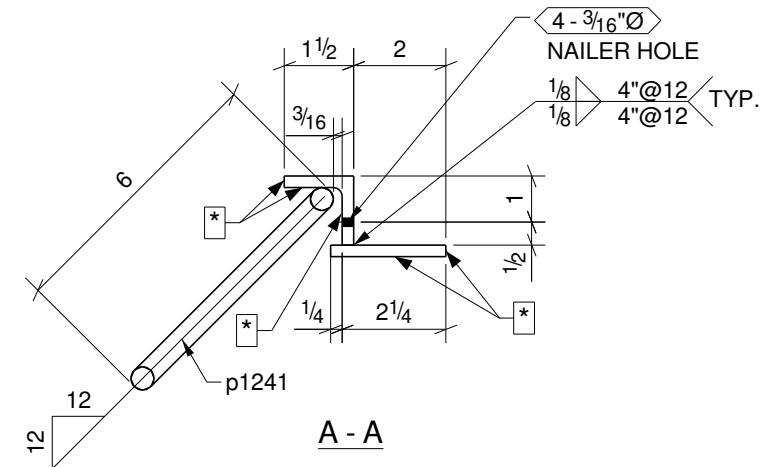
REF. DWGS.: 02-S-01-105



1~FRAME~02-1109

REF. E-DWGS. : 02-E139

MARK	QTY	DESCRIPTION	LENGTH		WEIGHT (LBS)	REMARKS
			FT	INS		
02-1109	ONE	FRAME				
02-1109	1	L1-1/2X1-1/2X1/4	5	1"	4.1	6061-T6
p1240	1	PL1/4"X2 1/2"	4	11"	3.5	6061-T6
p1241	12	RB1/2"	0	9"	1.9	6061-T6
p1242	1	L1-1/2X1-1/2X1/4	4	9 1/2"	3.9	6061-T6
p1243	2	PL1/4"X2 1/2"	4	8 1/2"	6.8	6061-T6
p1244	1	L1-1/2X1-1/2X1/4	4	9 1/2"	3.9	6061-T6



SHOP NOTE:

* APPLY THIS FACE WITH BITUMINUOS COAT

NOTES

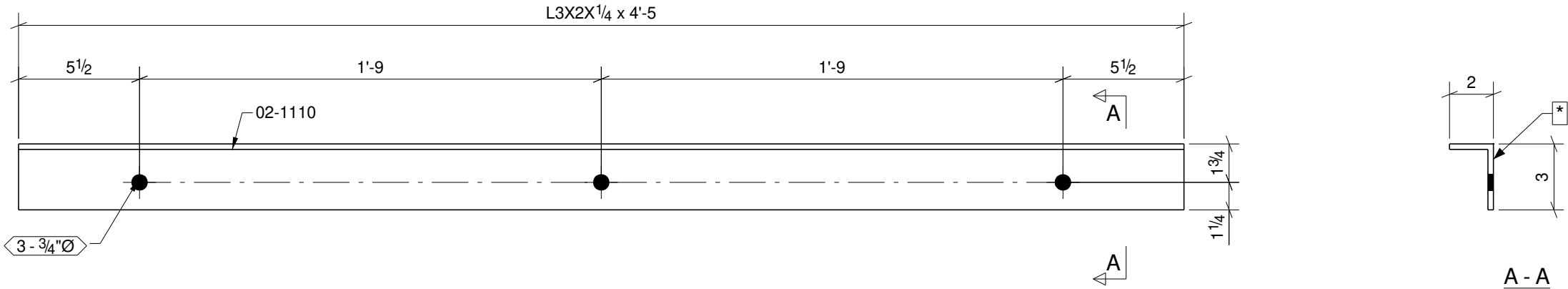
1. ALL MATERIAL TO MEET _____ AISC _____ SPECIFICATION UNLESS NOTED.
2. ALL WELDING ELECTRODES SHALL BE _____ SEE WPS _____ UNLESS NOTED.
3. DUPLICATION OF PIECE MARK INDICATES DUPLICATION OF WELDING.
4. EDGE DISTANCE 1 1/4" UNLESS NOTED.
5. ALL MINOR MEMBERS ARE CENTERED ON MAJOR MEMBERS UNLESS NOTED.
6. SHIPPING MARK SHALL BE IN POSITION SHOWN ON ERECTION DIAGRAM.
7. BACK CHARGES FOR ANY CORRECTIVE WORK MUST BE APPROVED BY CODY BUILDERS SUPPLY BEFORE WORK IS BEGUN.
8. ALL STEEL SHALL BE FABRICATED IN ACCORDANCE WITH LATEST A.I.S.C. SPECIFICATIONS ALUMINUM AND STAINLESS STEEL SHALL BE FABRICATED IN ACCORDANCE WITH LATEST N.A.A.M.M. SPECIFICATIONS.
9. ☒ -DENOTES TAIL DIMENSION STARTS.

BOI DENOTES : BOUGHT OUT ITEM

DETAILER	DUS	CHECKER	DVO
WELDS	1/4 U.N.O	HOLES	13/16 Ø U.N.O
SURF PREP : MILL FINISH			
PAINT : NONE			
REVISIONS			
No	DATE	BY	DESCRIPTION
A	06/30/2025	DUS	FOR APPROVAL

 SPECIALTY METALS FABRICATION 12002 N. LAMAR BLVD. AUSTIN, TEXAS 78753 WWW.CODYBUILDERSSUPPLY.COM (512) 339-9834			DETAILS OF	FRAME	JOB No.
			CUSTOMER	COUNTRY OF HAWAII	23240832
			PROJECT	HILO WWTP REHABILITATION	DRAWING No.
			LOCATION	HILO, HAWAII	02-1109
			ENGINEER	BRIAN F. FUNAI	

MARK	QTY	DESCRIPTION	LENGTH		WEIGHT	REMARKS	
			FT	INS	(LBS)		
02-1110	ONE	ANGLE					
02-1110	1	L3X2X1/4	4	5"	6.2	6061-T6	
FIELD BOLTS							
	3	5/8"Ø SS CONC. ANC BOLTS			0	4 1/2"	W/1 NUT & WASHER



1~ANGLE~02-1110

SHOP NOTE:
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NOTES

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9. ☒ -DENOTES TAIL DIMENSION STARTS.

BOI DENOTES : BOUGHT OUT ITEM

DETAILER	DUS	CHECKER	DVO
WELDS	1/4 U.N.O	HOLES	13/16 Ø U.N.O
SURF PREP : MILL FINISH			
PAINT : NONE			
REVISIONS			
No	DATE	BY	DESCRIPTION
A	06/30/2025	DUS	FOR APPROVAL

SPECIALTY METALS FABRICATION
12002 N. LAMAR BLVD, AUSTIN, TEXAS 78753
WWW.CODYBUILDERSSUPPLY.COM
(512) 339-9834

DETAILS OF	ANGLE	JOB No.
CUSTOMER	COUNTRY OF HAWAII	23240832
PROJECT	HILO WWTP REHABILITATION	DRAWING No.
LOCATION	HILO, HAWAII	02-1110
ENGINEER	BRIAN F. FUNAI	